

# **ME 165: Basic Mechanical Engineering**

## **Robotics**

### **Motion Characteristics**



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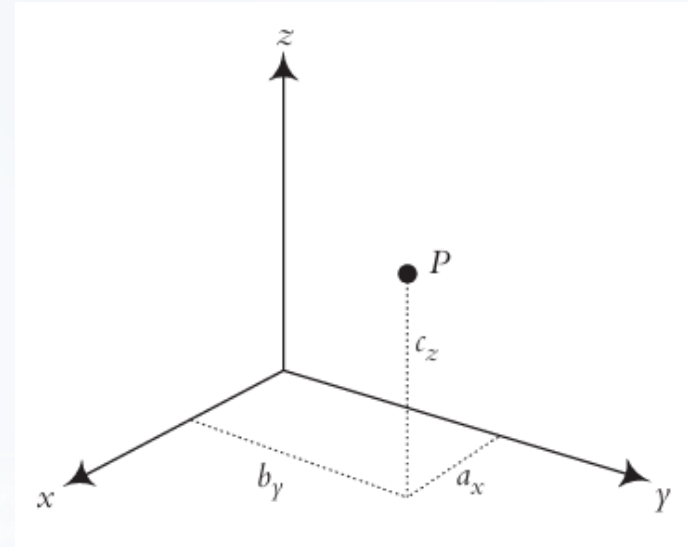
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# Matrix Representation

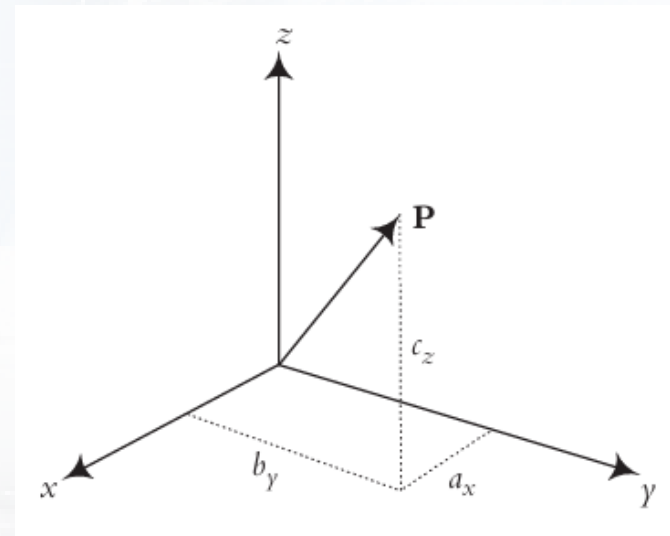
## ❖ Representation of a Point in Space

$$P = a_x \mathbf{i} + b_y \mathbf{j} + c_z \mathbf{k}$$



## ❖ Representation of a Vector in Space

$$P = a_x \mathbf{i} + b_y \mathbf{j} + c_z \mathbf{k}$$





# Matrix Representation

$$\mathbf{P} = \begin{bmatrix} a_x \\ b_y \\ c_z \end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix} P_x \\ P_y \\ P_z \\ w \end{bmatrix} \quad \text{where } a_x = \frac{P_x}{w}, b_y = \frac{P_y}{w}, \text{ etc.}$$

What happen if **w = 0 and 1?**



## Example:

A vector is described as  $\mathbf{P} = 3\mathbf{i} + 5\mathbf{j} + 2\mathbf{k}$ . Express the vector in matrix form:

- (a) With a scale factor of 2.
- (b) If it were to describe a direction as a unit vector.

## Solution:

$$\mathbf{P} = \begin{bmatrix} 6 \\ 10 \\ 4 \\ 2 \end{bmatrix}$$

$$\mathbf{P}_{unit} = \begin{bmatrix} 0.487 \\ 0.811 \\ 0.324 \\ 0 \end{bmatrix}$$

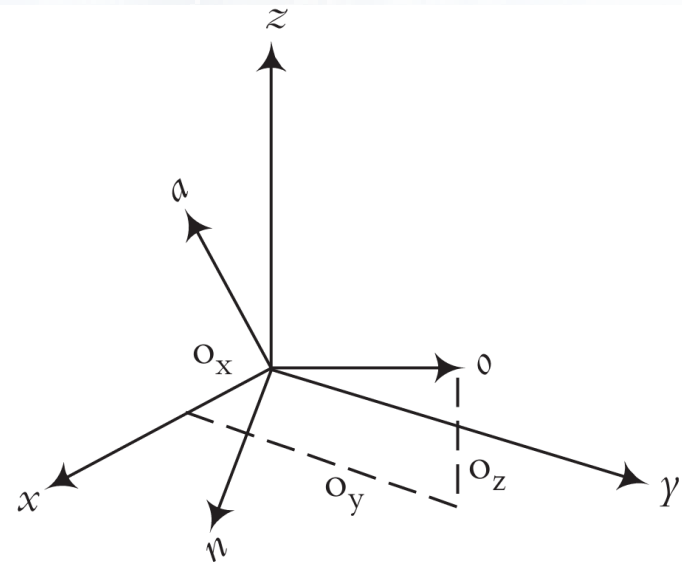


# Matrix Representation

## ❖ Representation of a Frame at the Origin of a Fixed Reference Frame

A frame is generally represented by three mutually orthogonal axes (such as  $x$ ,  $y$ , and  $z$ ). Since we may have more than one frame at any given time, we will use axes  $x$ ,  $y$ , and  $z$  to represent the fixed **Universe reference frame**  $F_{x,y,z}$  and a set of axes  $n$ ,  $o$ , and  $a$  to represent another **(moving) frame**  $F_{n,o,a}$  relative to the **reference frame**. This way, there should be no confusion about which frame is referenced.

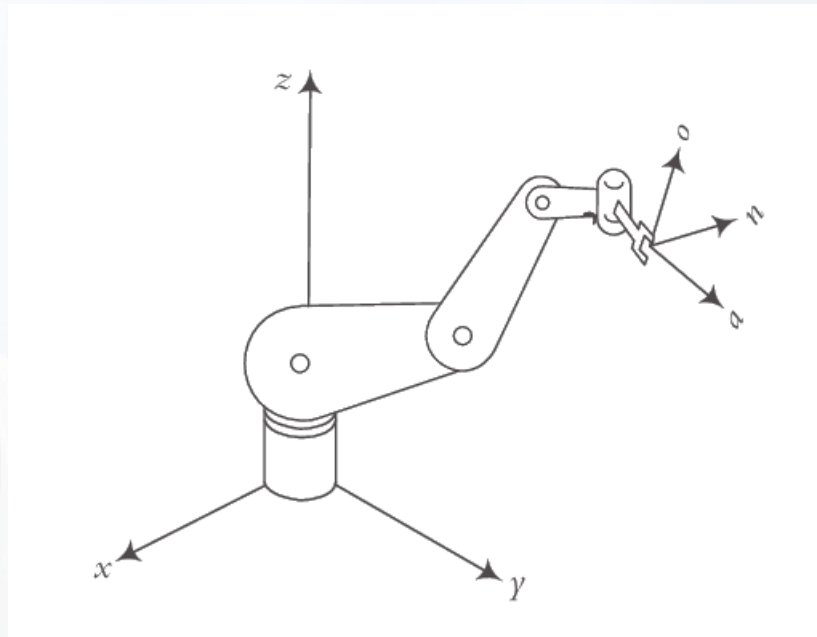
$$F = \begin{bmatrix} n_x & o_x & a_x \\ n_y & o_y & a_y \\ n_z & o_z & a_z \end{bmatrix}$$





# Matrix Representation

- $n$ ,  $o$ , and  $a$  are derived from the words **normal**, **orientation**, and **approach**



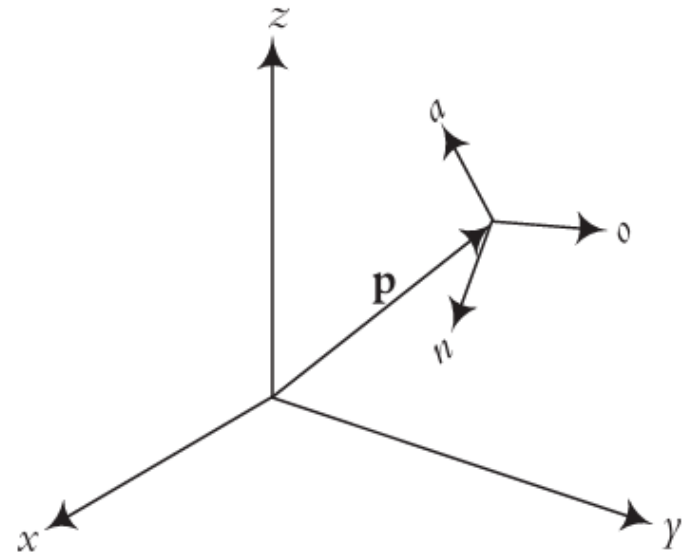


# Matrix Representation

## ❖ Representation of a Frame Relative to a Fixed Reference Frame

A frame may also be represented by a  $3 \times 4$  matrix without the scale factors, but it is not common. Adding the fourth row of scale factors to the matrix makes it a  $4 \times 4$  or *homogeneous* matrix.

$$F = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



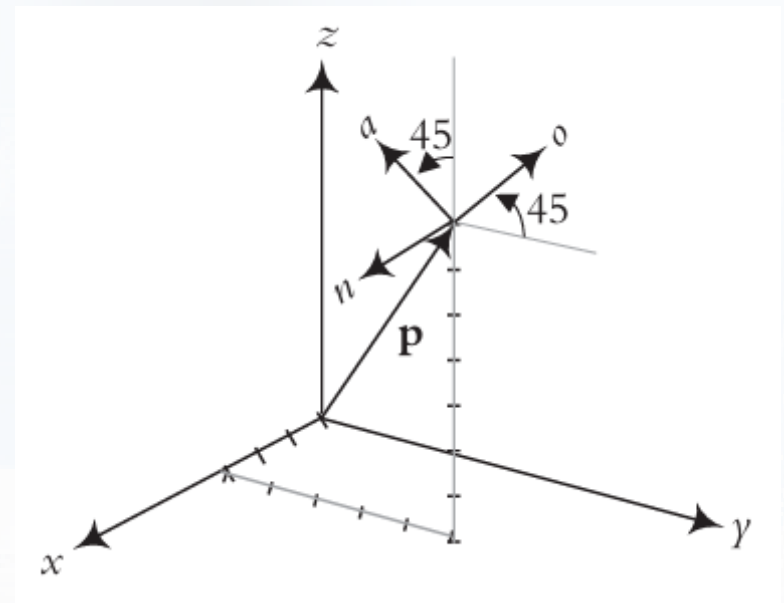


# Matrix Representation

## Example:

A frame  $F$  is located at 3, 5, 7 units, with its  $n$ -axis parallel to  $x$ , its  $o$ -axis at  $45^\circ$  relative to the  $y$ -axis, and its  $a$ -axis at  $45^\circ$  relative to the  $z$ -axis. Describe the frame.

$$F = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 0.707 & -0.707 & 5 \\ 0 & 0.707 & 0.707 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$





# Representation of Transformations



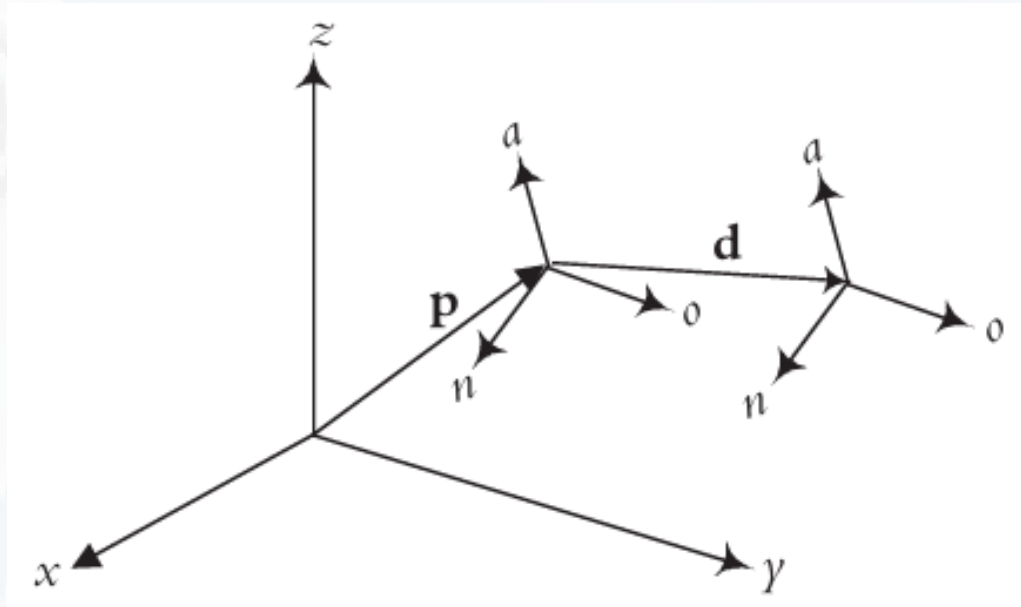
**A transformation is defined as making a movement in space**

A transformation may be in one of the following forms:

- **A pure translation**
- **A pure rotation about an axis**
- **A combination of translations and/or rotations**



# Representation of Pure Transformations



$$T = \begin{bmatrix} 1 & 0 & 0 & d_x \\ 0 & 1 & 0 & d_y \\ 0 & 0 & 1 & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



# Representation of Pure Transformations

## New Location of the Frame:

$$F_{new} = \begin{bmatrix} 1 & 0 & 0 & d_x \\ 0 & 1 & 0 & d_y \\ 0 & 0 & 1 & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} n_x & o_x & a_x & p_x + d_x \\ n_y & o_y & a_y & p_y + d_y \\ n_z & o_z & a_z & p_z + d_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This equation is also symbolically written as:

$$F_{new} = Trans(d_x, d_y, d_z) \times F_{old}$$



# Representation of Pure Transformations

## Example:

A frame  $F$  has been moved 10 units along the  $y$ -axis and 5 units along the  $z$ -axis of the reference frame. Find the new location of the frame.

$$F = \begin{bmatrix} 0.527 & -0.574 & 0.628 & 5 \\ 0.369 & 0.819 & 0.439 & 3 \\ -0.766 & 0 & 0.643 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



# Representation of Pure Transformations

**Solution:**

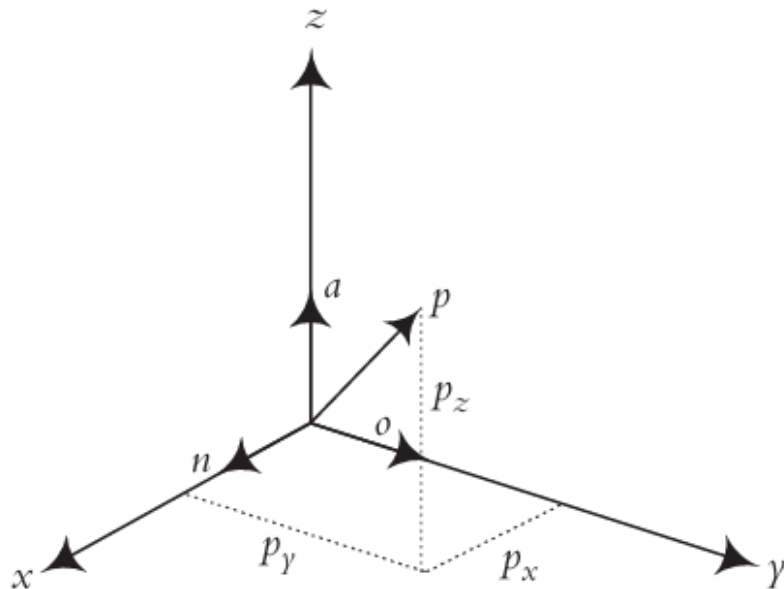
$$F_{new} = Trans(d_x, d_y, d_z) \times F_{old} = Trans(0, 10, 5) \times F_{old}$$

$$F_{new} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0.527 & -0.574 & 0.628 & 5 \\ 0.369 & 0.819 & 0.439 & 3 \\ -0.766 & 0 & 0.643 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 0.527 & -0.574 & 0.628 & 5 \\ 0.369 & 0.819 & 0.439 & 13 \\ -0.766 & 0 & 0.643 & 13 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

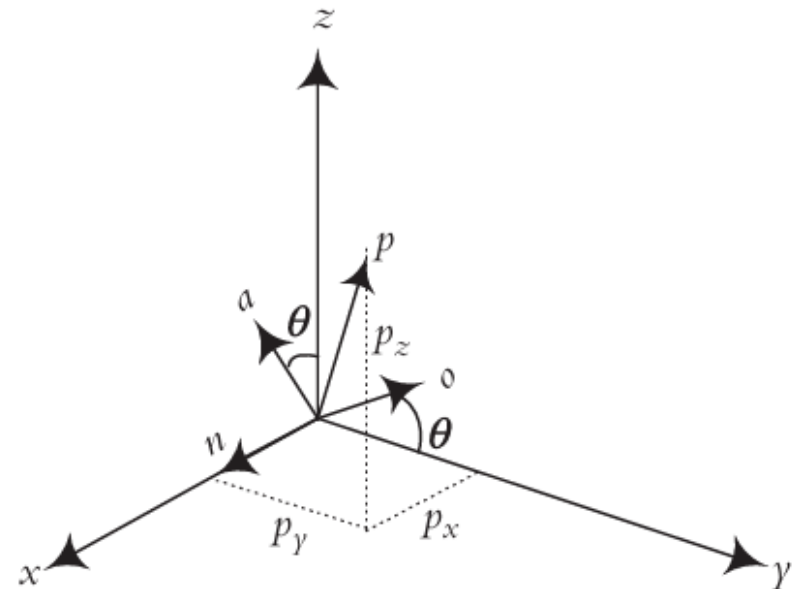


# Representation of Pure Rotation about an Axis

Assume a frame  $F_{noa'}$ , located at the origin of the reference frame  $F_{xyz}$ , rotates an **angle of  $\theta$**  about the  **$x$ -axis** of the reference frame.



Before rotation



After rotation





# Representation of Pure Rotation about an Axis

$$\begin{aligned} p_x &= p_n \\ p_y &= l_1 - l_2 = p_o \cos \theta - p_a \sin \theta \\ p_z &= l_3 + l_4 = p_o \sin \theta + p_a \cos \theta \end{aligned}$$

$$\begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} p_n \\ p_o \\ p_a \end{bmatrix}$$

Or,

$$p_{xyz} = Rot(x, \theta) \times p_{noa}$$

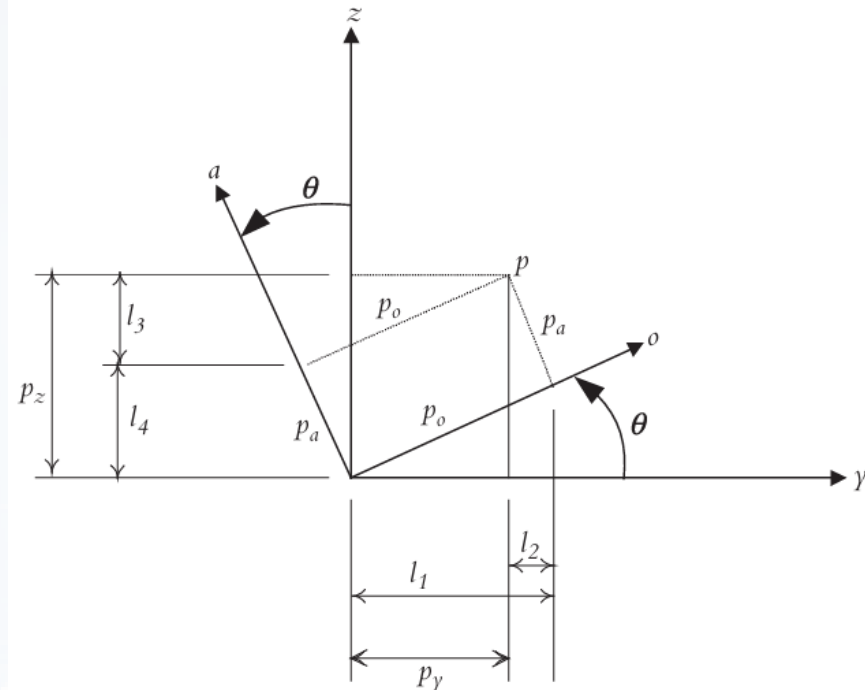
Or,

$${}^U p = {}^U T_R \times {}^R p$$

${}^U T_R$  = Transformation of frame  $R$  relative to frame  $U$

${}^U p = p_{xyz}$  ( $p$  relative to frame  $U$ )

${}^R p = p_{noa}$  ( $p$  relative to frame  $R$ )







# Representation of Pure Rotation about an Axis

$$Rot(x, \theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & C\theta & -S\theta \\ 0 & S\theta & C\theta \end{bmatrix}$$

$$Rot(y, \theta) = \begin{bmatrix} C\theta & 0 & S\theta \\ 0 & 1 & 0 \\ -S\theta & 0 & C\theta \end{bmatrix}$$

$$Rot(z, \theta) = \begin{bmatrix} C\theta & -S\theta & 0 \\ S\theta & C\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



# Representation of Pure Rotation about an Axis

## Example:

A point  $p(2,3,4)^T$  is attached to a rotating frame. The frame rotates  $90^\circ$  about the  $x$ -axis of the reference frame. Find the coordinates of the point relative to the reference frame after the rotation, and verify the result graphically.

## Hints:

$$Rot(x, \theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & C\theta & -S\theta \\ 0 & S\theta & C\theta \end{bmatrix}$$

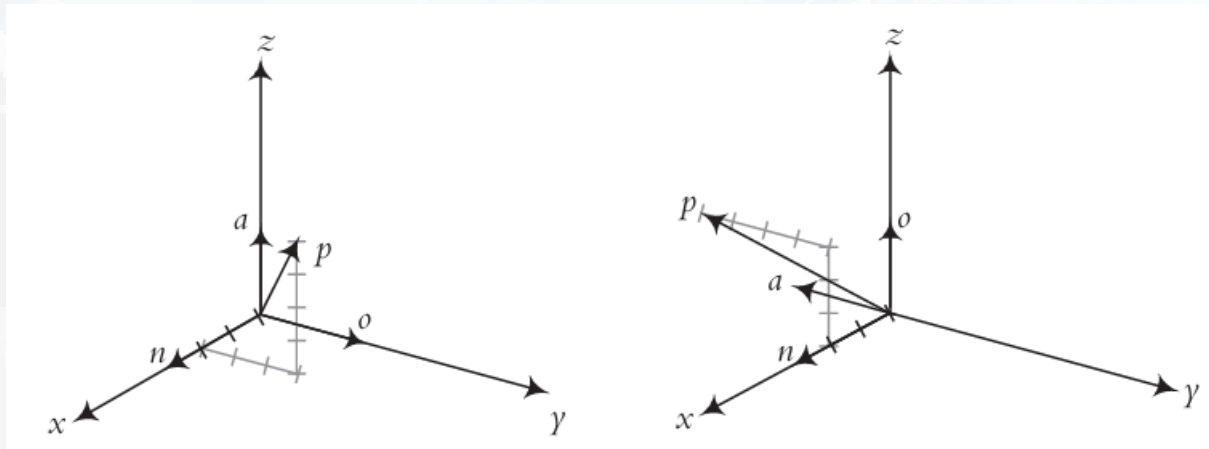


# Representation of Pure Rotation about an Axis

**Solution:**

$$\begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & C\theta & -S\theta \\ 0 & S\theta & C\theta \end{bmatrix} \times \begin{bmatrix} p_n \\ p_o \\ p_a \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 3 \end{bmatrix}$$





# Combined Transformations

Let's assume that a frame  $F_{noa}$  is subjected to the following three successive transformations relative to the reference frame  $F_{xyz}$ :

1. Rotation of  $\alpha$  degrees about the x-axis,
2. Followed by a translation of  $[l_1, l_2, l_3]$  (relative to the x-, y-, and z-axes respectively),
3. Followed by a rotation of  $\beta$  degrees about the y-axis.

1. 
$$p_{1,xyz} = Rot(x, \alpha) \times p_{noa}$$

2. 
$$p_{2,xyz} = Trans(l_1, l_2, l_3) \times p_{1,xyz} = Trans(l_1, l_2, l_3) \times Rot(x, \alpha) \times p_{noa}$$

3. 
$$p_{xyz} = p_{3,xyz} = Rot(y, \beta) \times p_{2,xyz} = Rot(y, \beta) \times Trans(l_1, l_2, l_3) \times Rot(x, \alpha) \times p_{noa}$$



# Combined Transformations

## Example:

A point  $p(7,3,1)^T$  is attached to a frame  $F_{noa}$  and is subjected to the following transformations. Find the coordinates of the point relative to the reference frame at the conclusion of transformations.

1. Rotation of  $90^\circ$  about the  $z$ -axis,
2. Followed by a rotation of  $90^\circ$  about the  $y$ -axis,
3. Followed by a translation of  $[4, -3, 7]$ .

**Solution:** The matrix equation representing the transformation is:

$$p_{xyz} = Trans(4, -3, 7)Rot(y, 90)Rot(z, 90)p_{noa}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 7 \\ 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 10 \\ 1 \end{bmatrix}$$

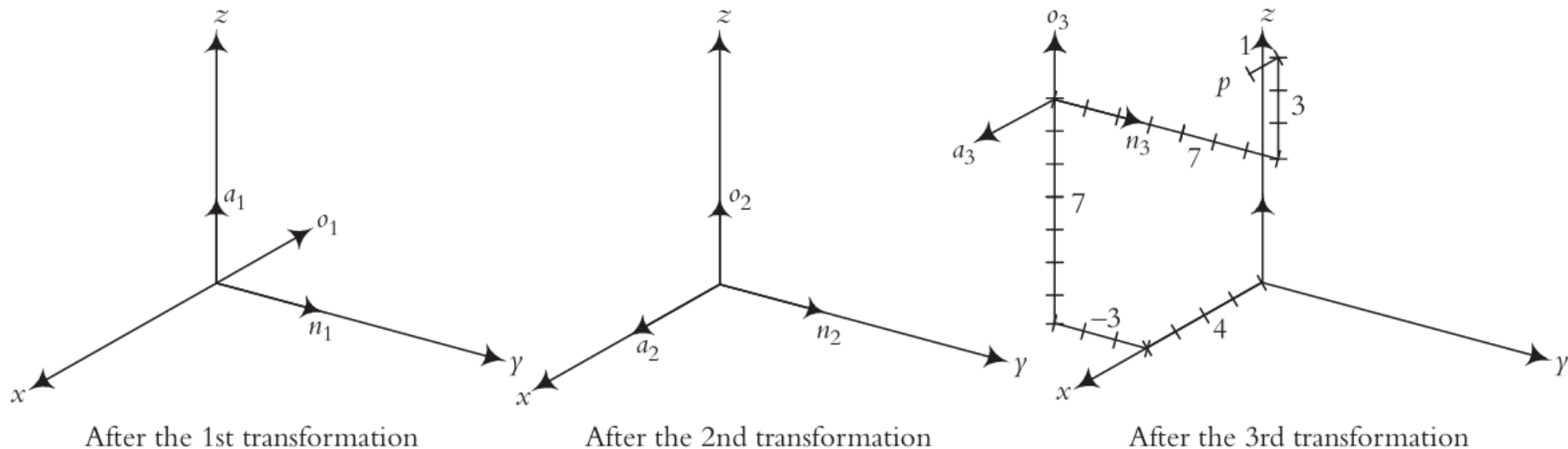


# Combined Transformations

## Example:

A point  $p(7,3,1)^T$  is attached to a frame  $F_{noa}$  and is subjected to the following transformations. Find the coordinates of the point relative to the reference frame at the conclusion of transformations.

1. Rotation of  $90^\circ$  about the  $z$ -axis,
2. Followed by a rotation of  $90^\circ$  about the  $y$ -axis,
3. Followed by a translation of  $[4, -3, 7]$ .





# Combined Transformations

## Example:

In this case, assume the same point  $p(7,3,1)^T$ , attached to  $F_{noa}$ , is subjected to the same transformations, but the transformations are performed in a different order, as shown. Find the coordinates of the point relative to the reference frame at the conclusion of transformations.

1. A rotation of  $90^\circ$  about the  $z$ -axis,
2. Followed by a translation of  $[4, -3, 7]$ ,
3. Followed by a rotation of  $90^\circ$  about the  $y$ -axis.

**Solution:** The matrix equation representing the transformation is:

$$p_{xyz} = Rot(y, 90) Trans(4, -3, 7) Rot(z, 90) p_{noa}$$
$$= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 7 \\ 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \\ -1 \\ 1 \end{bmatrix}$$



# Combined Transformations

## Example:

Assume that the same point as in Previous Example is now subjected to the same transformations, but all relative to the current moving frame, as listed below. Find the coordinates of the point relative to the reference frame after transformations are completed.

1. A rotation of  $90^\circ$  about the  $a$ -axis,
2. Then a translation of  $[4, -3, 7]$  along  $n$ -,  $o$ -,  $a$ -axes
3. Followed by a rotation of  $90^\circ$  about the  $o$ -axis.

## Solution:

$$p_{xyz} = Rot(a, 90) Trans(4, -3, 7) Rot(o, 90) p_{noa}$$

$$= \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 7 \\ 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 0 \\ 1 \end{bmatrix}$$





# Combined Transformations

## Example:

A frame  $B$  was rotated about the  $x$ -axis  $90^\circ$ , then it was translated about the current  $a$ -axis 3 inches before it was rotated about the  $z$ -axis  $90^\circ$ . Finally, it was translated about current  $o$ -axis 5 inches.

- (a) Write an equation that describes the motions.
- (b) Find the final location of a point  $p(1,5,4)^T$  attached to the frame relative to the reference frame.

**Solution:** In this case, motions alternate relative to the reference frame and current frame.

- (a) Pre- or post-multiplying each motion's matrix accordingly, we will get:

$${}^U T_B = Rot(z, 90) Rot(x, 90) Trans(0, 0, 3) Trans(0, 5, 0)$$



# Combined Transformations

## Example:

A frame  $B$  was rotated about the  $x$ -axis  $90^\circ$ , then it was translated about the current  $a$ -axis 3 inches before it was rotated about the  $z$ -axis  $90^\circ$ . Finally, it was translated about current  $o$ -axis 5 inches.

- (a) Write an equation that describes the motions.
- (b) Find the final location of a point  $p(1,5,4)^T$  attached to the frame relative to the reference frame.

$${}^U T_B = Rot(z, 90) Rot(x, 90) Trans(0, 0, 3) Trans(0, 5, 0)$$

- (b) Substituting the matrices and multiplying them, we will get:

$${}^U p = {}^U T_B \times {}^B p$$

$$= \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \\ 10 \\ 1 \end{bmatrix}$$



# Combined Transformations

## Example:

A frame  $F$  was rotated about the  $y$ -axis  $90^\circ$ , followed by a rotation about the  $o$ -axis of  $30^\circ$ , followed by a translation of 5 units along the  $n$ -axis, and finally, a translation of 4 units along the  $x$ -axis. Find the total transformation matrix.

**Solution:** The following set of matrices, written in the proper order to represent transformations relative to the reference frame or the current frame describes the total transformation:

$$T = Trans(4, 0, 0)Rot(y, 90)Rot(o, 30)Trans(5, 0, 0)$$

$$= \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0.866 & 0 & 0.5 & 0 \\ 0 & 1 & 0 & 0 \\ -0.5 & 0 & 0.866 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -0.5 & 0 & 0.866 & 1.5 \\ 0 & 1 & 0 & 0 \\ -0.866 & 0 & -0.5 & -4.33 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

# Forward and Inverse Kinematics



- If we have a robot whose configuration is known i.e., all the link lengths and joint angles are known. Calculating the position and orientation of the hand of the robot is called **forward kinematic analysis**.
- If we want to place the hand of the robot at a desired location and orientation, we need to know how much each link length or joint angle must be such that at those values the hand will be at the desired position and orientation. This is called **inverse kinematic analysis**.
- In reality, the **inverse kinematic analysis is far more important** to enable the robot controller to calculate the joint values using these equations and to run the robot to the desired position and orientation.

# Forward and Inverse Kinematics



The following possibilities will be discussed:

(a) Cartesian (gantry, rectangular) coordinates

(b) Cylindrical coordinates

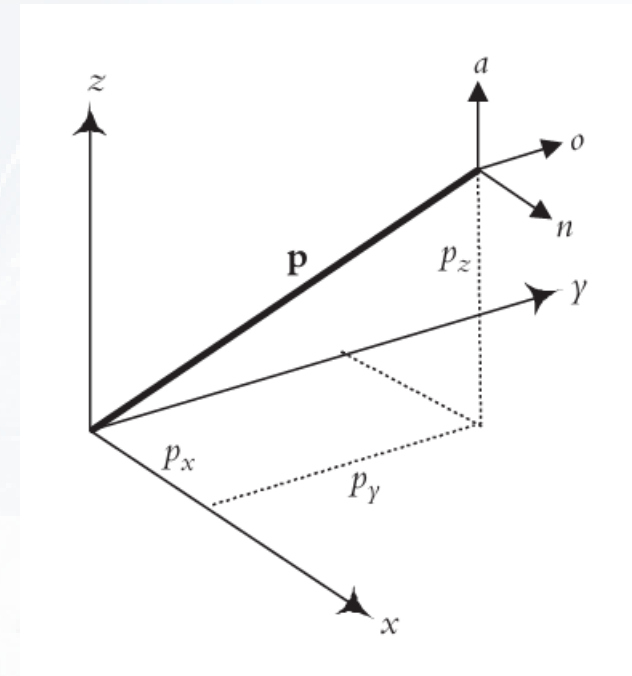
(c) Spherical coordinates



# Cartesian Robot

In a cartesian robot, the transformation matrix representing this motion to point  $P$  is a **simple translation matrix**. The transformation matrix representing the forward kinematic equation of the position of the hand of the robot in a Cartesian coordinate system will be:

$${}^R T_P = T_{cart}(p_x, p_y, p_z) = \begin{bmatrix} 1 & 0 & 0 & p_x \\ 0 & 1 & 0 & p_y \\ 0 & 0 & 1 & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$





## Cartesian Robot

### Example:

It is desired to position the origin of the hand frame of a Cartesian robot at point  $p = [3 \ 4 \ 7]^T$ . Calculate the necessary Cartesian coordinate motions that need to be made.

### Solution:

Setting the forward kinematic equation, represented by the  ${}^R T_p$  matrix, equal to the desired position yields:

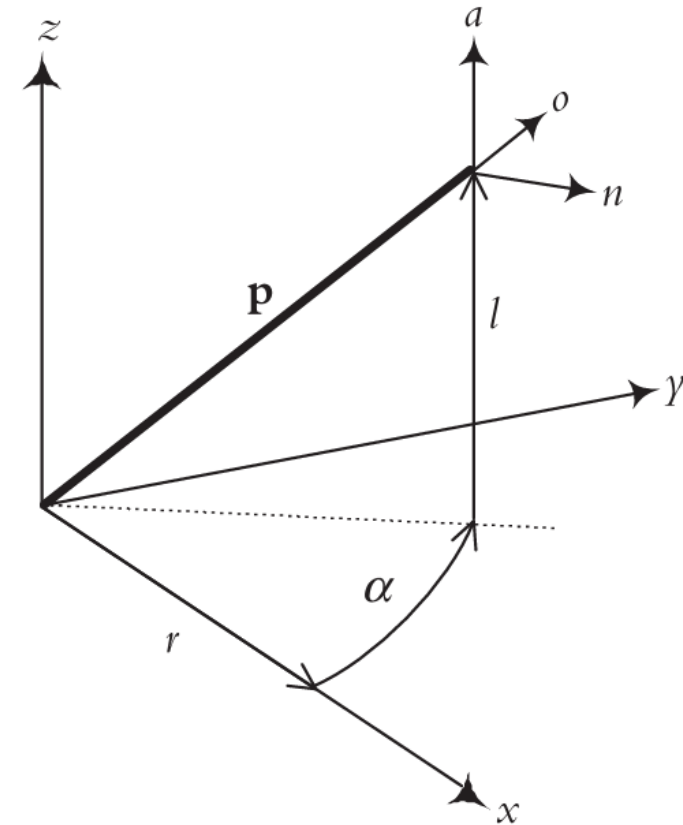
$${}^R T_p = \begin{bmatrix} 1 & 0 & 0 & p_x \\ 0 & 1 & 0 & p_y \\ 0 & 0 & 1 & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{or } p_x = 3, p_y = 4, p_z = 7$$



# Cylindrical Robot

A cylindrical coordinate system includes **two linear translations and one rotation**. The sequence is a **translation of  $r$  along the  $x$ -axis**, a **rotation of  $\alpha$  about the  $z$ -axis**, and a **translation of  $l$  along the  $z$ -axis**. Since these transformations are all relative to the Universe frame, the total transformation caused by these three transformations is found by pre-multiplying each matrix, as follows:



$${}^R T_p = T_{cyl}(r, \alpha, l) = Trans(0, 0, l) Rot(z, \alpha) Trans(r, 0, 0)$$





# Cylindrical Robot

$${}^R T_p = T_{cyl}(r, \alpha, l) = Trans(0, 0, l) Rot(z, \alpha) Trans(r, 0, 0)$$

$${}^R T_p = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & l \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} C\alpha & -S\alpha & 0 & 0 \\ S\alpha & C\alpha & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & r \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^R T_p = T_{cyl}(r, \alpha, l) = \begin{bmatrix} C\alpha & -S\alpha & 0 & rC\alpha \\ S\alpha & C\alpha & 0 & rS\alpha \\ 0 & 0 & 1 & l \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



# Cylindrical Robot

The original orientation of the frame may be restored by rotating the  $n, o, a$  frame about the  $a$ -axis at an angle of  $-\alpha$ , which is equivalent to post-multiplying the cylindrical coordinate matrix by a rotation matrix of  $\text{Rot}(a, -\alpha)$ . As a result, the frame will be at the same location but will be parallel to the reference frame again, as follows:

$$\begin{aligned}
 T_{cyl} \times \text{Rot}(a, -\alpha) &= \begin{bmatrix} C\alpha & -S\alpha & 0 & rC\alpha \\ S\alpha & C\alpha & 0 & rS\alpha \\ 0 & 0 & 1 & l \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} C(-\alpha) & -S(-\alpha) & 0 & 0 \\ S(-\alpha) & C(-\alpha) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 & rC\alpha \\ 0 & 1 & 0 & rS\alpha \\ 0 & 0 & 1 & l \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$



# Cylindrical Robot

## Example:

It is desired to place the origin of the hand frame of a Cylindrical robot at point  $p = [3 \ 4 \ 7]^T$ . Calculate the joint variables of the robot.

**Solution:** Setting the components of the location of the origin of the frame from the  $T_{cyl}$  matrix we get:

$$l = 7$$

$$rC\alpha = 3 \quad \text{and} \quad rS\alpha = 4 \quad \text{and therefore, } \tan \alpha = 4/3 \text{ and } \alpha = 53.1^\circ$$

Substituting  $\alpha$  into either equation will yield  $r = 5$ . The final answer is  $r = 5$  units,  $\alpha = 53.1^\circ$ , and  $l = 7$  units.



# Cylindrical Robot

## Example:

The position and restored orientation of a cylindrical robot are given. Find the matrix representing the original position and orientation of the robot before it was restored.

$$T = \begin{bmatrix} 1 & 0 & 0 & -2.394 \\ 0 & 1 & 0 & 6.578 \\ 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



# Cylindrical Robot

**Solution:** Since  $r$  is always positive, it is clear that  $S\alpha$  and  $C\alpha$  are positive and negative, respectively. Therefore,  $\alpha$  is in the second quadrant. From  $T$ , we get:

$$l = 9$$

$$\tan(\alpha) = \frac{6.578}{-2.394} = -2.748 \quad \rightarrow \quad \alpha = 180^\circ - 70^\circ = 110^\circ$$

$$r \sin(\alpha) = 6.578 \rightarrow r = 7$$

Substituting these values into Equation (2.34), we will get the original orientation of the robot:

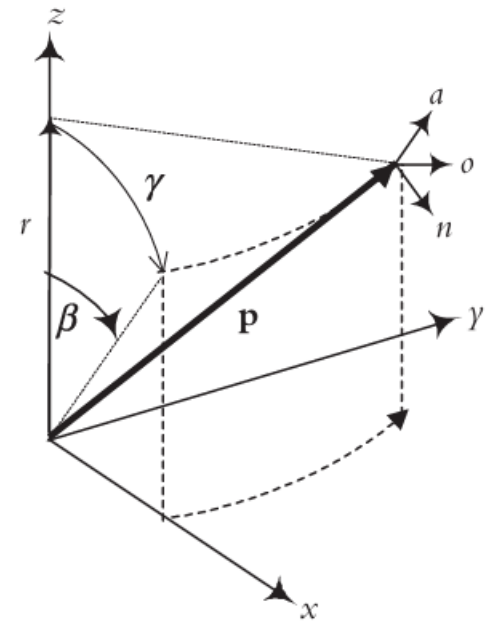
$${}^R T_p = \begin{bmatrix} C\alpha & -S\alpha & 0 & rC\alpha \\ S\alpha & C\alpha & 0 & rS\alpha \\ 0 & 0 & 1 & l \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -0.342 & -0.9397 & 0 & -2.394 \\ 0.9397 & -0.342 & 0 & 6.578 \\ 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \blacksquare$$



# Spherical Robot

A spherical coordinate system consists of one linear motion and two rotations. The sequence is a translation of  $r$  along the  $z$ -axis, a rotation of  $\beta$  about the  $y$ -axis, and a rotation of  $\gamma$  about the  $z$ -axis as shown in Figure . Since these transformations are all relative to the Universe frame, the total transformation caused by these three transformations can be found by pre-multiplying by each matrix, as follows:

$${}^R T_P = T_{sph}(r, \beta, \gamma) = Rot(z, \gamma) Rot(y, \beta) Trans(0, 0, r)$$





# Spherical Robot

$${}^R T_P = T_{sph}(r, \beta, \gamma) = Rot(z, \gamma) Rot(y, \beta) Trans(0, 0, r)$$

$${}^R T_P = \begin{bmatrix} C\gamma & -S\gamma & 0 & 0 \\ S\gamma & C\gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} C\beta & 0 & S\beta & 0 \\ 0 & 1 & 0 & 0 \\ -S\beta & 0 & C\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & r \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^R T_P = T_{sph}(r, \beta, \gamma) = \begin{bmatrix} C\beta C\gamma & -S\gamma & S\beta C\gamma & rS\beta C\gamma \\ C\beta S\gamma & C\gamma & S\beta S\gamma & rS\beta S\gamma \\ -S\beta & 0 & C\beta & rC\beta \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



# Spherical Robot

## Example:

Suppose we now want to place the origin of the hand of a spherical robot at  $[3,4,7]^T$ . Calculate the joint variables of the robot.

**Solution:** Setting the components of the location of the origin of the frame from  $T_{sph}$  matrix of Equation (2.36) to the desired values, we get:

$$rS\beta C\gamma = 3$$

$$rS\beta S\gamma = 4$$

$$rC\beta = 7$$

From the third equation, we determine that the  $C\beta$  is positive, but there is no such information about  $S\beta$ . Therefore, because we do not know the actual sign of  $S\beta$ , there are two possible solutions. Later, we will have to check the final results to ensure they are correct.

$$\tan\gamma = 4/3 \rightarrow \gamma = 53.1^\circ \quad \text{or} \quad 233.1^\circ$$

$$\text{then} \quad S\gamma = 0.8 \quad \text{or} \quad -0.8$$

$$\text{and} \quad C\gamma = 0.6 \quad \text{or} \quad -0.6$$

$$\text{and} \quad rS\beta = 3/0.6 = 5 \quad \text{or} \quad -5$$

$$\text{and since} \quad rC\beta = 7, \rightarrow \beta = 35.5^\circ \quad \text{or} \quad -35.5^\circ$$

$$\text{and} \quad r = 8.6$$





# Spherical Robot

## Example:

$$\begin{array}{llll}
 \tan \gamma = 4/3 & \rightarrow & \gamma = 53.1^\circ & \text{or } 233.1^\circ \\
 \text{then} & & S\gamma = 0.8 & \text{or } -0.8 \\
 \text{and} & & C\gamma = 0.6 & \text{or } -0.6 \\
 \text{and} & & rS\beta = 3/0.6 = 5 & \text{or } -5 \\
 \text{and since} & & rC\beta = 7, & \rightarrow \beta = 35.5^\circ \text{ or } -35.5^\circ \\
 \text{and} & & r = 8.6 & 
 \end{array}$$

You may check both answers and verify that they **both satisfy all position equations**. If you also follow these angles about the given axes in 3D, you get to the same point physically. But these two answers result in the **same position, but at different orientations**. So, the orientation terms in the transformation matrix will be different for the two answers. Since we are not concerned with the orientation of the hand frame at this point, both position answers are correct.



Thank You